

## MindMatrix VTU Internship Program

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Project Title: 99

### Android App Development using GenAI Shilpa-Kala Showcase (National Pride)

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|----------------|-----------------------------------------------------------------------------|
| Project Title  | Android App Development using GenAI – Shilpa-Kala Showcase (National Pride) |
| Project Number | 99                                                                          |
| Platform       | Android (Kotlin + XML Layouts)                                              |
| Domain         | Cultural Heritage / Art Commerce / National Pride                           |
| Category       | Android Application with GenAI Integration                                  |
| Report Type    | Internship / Academic Project Report                                        |

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## 1. Introduction

Shilpa-Kala Showcase is an Android application developed to bridge the critical digital divide between India's master stone and wood carvers and the modern global marketplace. The application name combines the Sanskrit word "Shilpa" (art/craft) with "Kala" (skill/artistry), reflecting its core purpose: to be the definitive digital portfolio platform for every traditional craftsman working in the shadows of India's rich sculptural heritage.

Developed using Kotlin with XML-based layouts, the app integrates the Glide image library for high-performance photo loading, Android Intents for seamless WhatsApp integration, and GenAI-assisted content generation for heritage storytelling. The application targets master carvers — particularly those practising ancient styles such as Hoysala, Dravidian, and Mysore stone carving — a segment that has historically relied on word-of-mouth, physical exhibitions, and intermediaries to reach buyers.

The application addresses three critical needs of traditional craftsmen: showcasing their portfolio of finished and in-progress works through a high-resolution digital gallery, enabling direct buyer contact through an inquiry mechanism that leverages the ubiquitous WhatsApp platform, and preserving cultural heritage narratives through AI-generated heritage storytelling sections that document rare carving techniques for posterity.

Shilpa-Kala Showcase is designed to function as a premium art gallery experience on any Android device. The UI philosophy is deliberately minimalist and gallery-like, placing the artwork front and centre without visual clutter. The app targets the underserved community of rural artisans across Karnataka, Rajasthan, Odisha, and Tamil Nadu who possess world-class craftsmanship but have no digital footprint with which to reach national and international buyers.

## 2. Problem Statement

India's stone and wood carving tradition is among the oldest and most sophisticated in the world. Craftsmen (Shilpis) in towns like Shivarapatna, Mahabalipuram, Puri, and Jaisalmer create sculptures of extraordinary quality — yet they remain almost entirely invisible to the national and global market. Several systemic problems perpetuate this invisibility:

| Challenge                        | Description                                                                                                                                                      |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>No Digital Presence</b>       | Skilled Shilpis have no website, social media profile, or online portfolio. International and national buyers cannot discover them through any digital channel.  |
| <b>Intermediary Dependency</b>   | Artisans rely on middlemen, exporters, and temple committees to sell their work, surrendering a disproportionate share of the value they create.                 |
| <b>Art Slowly Fading</b>         | The younger generation sees no modern career path in traditional carving. Without economic viability, the craft is abandoned for urban employment opportunities. |
| <b>No Inquiry Mechanism</b>      | Even when a buyer discovers an artisan's work, there is no structured way to make an inquiry or track a commission without exchanging physical contact details.  |
| <b>No Heritage Documentation</b> | Ancient carving styles and techniques are passed down orally and risk being permanently lost when a master craftsman retires without a documented record.        |
| <b>No Work-in-Progress View</b>  | Buyers cannot see the creation journey — a block of stone becoming a temple idol — which is itself a powerful value proposition for premium art buyers.          |

These interconnected challenges highlight the urgent need for a dedicated, beautifully designed digital platform that artisans can use with minimal technical knowledge, on affordable Android smartphones, to achieve global visibility for their work.

### 3. Project Vision

Shilpa-Kala Showcase envisions a world where every traditional Indian craftsman — regardless of education level, geographic location, or digital literacy — has access to a world-class digital gallery that presents their work with the dignity and professionalism it deserves, and connects them directly with buyers who value authentic handmade art.

The project is built on four foundational pillars:

- **Visibility** – Every artisan deserves a digital presence as polished as any premium art gallery. The app presents sculpture photographs with museum-quality framing, zoom capability, and curated descriptions.
- **Direct Connection** – By leveraging WhatsApp — already used by millions of rural Indians — the app enables buyers to send a pre-formatted inquiry directly to the artisan, eliminating the need for intermediaries.
- **Heritage Preservation** – GenAI-powered heritage storytelling sections document the history, techniques, and cultural significance of each carving style, creating a living digital record of India's sculptural traditions.
- **Economic Empowerment** – By giving artisans direct access to buyers, the app enables them to capture more value from their craft, making it economically viable for the next generation to continue the tradition.

Shilpa-Kala Showcase is not merely a portfolio app; it is a cultural preservation tool and an economic empowerment platform, designed to connect ancient art with the modern buyer one sculpture at a time.

## 4. Scope of the Application

### 4.1 Target Users

- **Primary:** Traditional stone and wood carvers practising Hoysala, Dravidian, Mysore, Rajasthani, and Orissan carving styles.
- **Secondary:** Art collectors, interior designers, temple committees, and cultural organisations seeking authentic handmade sculptures.
- **Tertiary:** NGOs, government cultural bodies, and exporters promoting India's handicraft heritage internationally.

### 4.2 Platform

- Android smartphones with minimum API Level 21 (Android 5.0 Lollipop and above).
- Optimised for mid-range and high-end devices to support high-resolution image rendering.
- Glide image library ensures smooth performance even on devices with limited RAM when handling 20+ high-resolution sculpture photographs.

### 4.3 Key Operational Scenarios

- Browsing an artist's complete portfolio of finished sculptures in a high-resolution gallery grid.
- Zooming into individual sculptures to inspect intricate carving details at full resolution.
- Viewing a work-in-progress timeline showing the transformation from a raw block of stone to a finished idol.
- Sending a pre-formatted WhatsApp inquiry message containing the product ID, artist name, and buyer's interest.
- Reading the heritage story of a specific carving style through an AI-generated narrative section.

## 5. Objectives

The project is aligned with the following measurable objectives:

- **Deliver a premium gallery experience:** Present 20 or more high-resolution sculpture photographs without frame drops, slow loads, or memory errors, using Glide's caching and bitmap pooling.
- **Enable detailed artwork inspection:** Implement a pinch-to-zoom and double-tap zoom feature on every gallery image so buyers can examine intricate carving details before making an inquiry.
- **Facilitate frictionless buyer inquiry:** Provide a single-tap Enquire button on every artwork that composes and launches a pre-formatted WhatsApp message containing the unique product ID and artwork details.
- **Document the creation journey:** Build a timeline / work-in-progress feature that displays a chronological series of photographs showing how a raw stone block becomes a finished sculpture.
- **Preserve cultural heritage narratives:** Integrate a heritage story section for each carving style (e.g., Hoysala, Dravidian) with AI-generated historical and cultural context.
- **Achieve a minimalist gallery-grade UI:** Design the interface to feel like a premium art gallery — clean backgrounds, generous whitespace, high-contrast typography, and artwork-first layout.
- **Support the Make in India vision:** Position the app as a platform for high-value export of traditional Indian crafts to international buyers.

## 6. Features

### 6.1 Core Features

| Feature                          | Description                                                                                                                                                                  |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Artist Portfolio Gallery</b>  | High-resolution image gallery showcasing completed sculptures. Glide loads and caches images smoothly. Grid view with artwork title, material, and dimensions.               |
| <b>Pinch-to-Zoom Viewer</b>      | Full-screen image viewer with pinch-to-zoom and double-tap zoom (up to 3x) allowing buyers to inspect intricate carving details at pixel level.                              |
| <b>Work-in-Progress Timeline</b> | Chronological timeline showing the step-by-step creation of a sculpture — raw stone, rough carving, fine detailing, polishing, and final idol — with captions at each stage. |
| <b>WhatsApp Order Inquiry</b>    | A single Enquire button on every artwork uses Android Intents to launch WhatsApp with a pre-formatted message containing product ID, artwork name, and artist details.       |
| <b>Heritage Story Section</b>    | AI-assisted narrative explaining the history, cultural significance, and technique of a specific carving style (e.g., Hoysala, Dravidian). Displayed with styled typography. |
| <b>Artwork Detail Screen</b>     | Dedicated screen for each sculpture showing full-size image, title, material (granite/sandstone/teak), dimensions, creation time, style category, and a short description.   |
| <b>Artist Profile Screen</b>     | Dedicated profile page for the Shilpi showing their name, location, years of experience, specialisation, and a brief biographical narrative.                                 |
| <b>Gallery Categories</b>        | Artworks organised by category: Deities, Decorative Panels, Architectural Elements, and Custom Commissions — enabling buyers to browse by type using filter chips.           |

### 6.2 Future Scope Features

- **Multi-Artist Platform:** Expand from a single-artist showcase to a marketplace featuring multiple verified Shilpis across India.
- **AR Preview:** Allow buyers to visualise a sculpture in their home or garden using Augmented Reality via ARCore.
- **Video Documentation:** Embed short video clips of the carving process alongside the photo timeline.
- **Multi-Language Support:** Extend the interface to Kannada, Hindi, Tamil, and Telugu for artisans with limited English literacy.
- **UPI / Razorpay Integration:** Enable advance booking payments directly within the app.
- **AI Price Estimator:** Use GenAI to generate price estimates based on sculpture dimensions, material, and complexity.

## 7. Functional Requirements

The following functional requirements define the core behaviours the system must support:

### FR-1: Gallery & Image Display

- The system shall display a scrollable grid of sculpture thumbnail images on the home screen.
- The system shall load images using Glide with placeholder and error drawables to handle failures gracefully.
- The system shall support a minimum of 20 high-resolution images without perceptible lag or Out-of-Memory errors.
- The system shall provide a full-screen image viewer launched by tapping any thumbnail.

### FR-2: Zoom and Interaction

- The full-screen viewer shall support pinch-to-zoom with a minimum zoom factor of 3x.
- The viewer shall support double-tap to toggle between fit-to-screen and 2x zoom.
- The viewer shall support drag-to-pan when the image is zoomed beyond the screen boundary.

### FR-3: Artwork Detail

- The system shall display a detail screen for each artwork containing: full-size image, title, material, dimensions, creation duration, style category, and a short description.
- The detail screen shall include a clearly visible Enquire via WhatsApp button.

### FR-4: WhatsApp Inquiry

- The system shall compose a pre-formatted WhatsApp message containing the product ID, artwork name, artist name, and a buyer message template.
- The system shall launch WhatsApp using an Android Intent with the ACTION\_VIEW URI scheme.
- The product ID must be unique per artwork and included verbatim in the WhatsApp message.

### FR-5: Work-in-Progress Timeline

- The system shall display a vertical timeline of photographs documenting the creation process of a selected sculpture.
- Each timeline entry shall include a photograph, a stage label (e.g., Day 1 - Stone Selection), and a brief caption.
- The timeline shall be scrollable and support the same zoom capability as the main gallery viewer.

### FR-6: Heritage Story

- The system shall display a heritage story screen accessible from the main navigation.



- The screen shall present AI-generated historical and cultural content about at least one carving style (e.g., Hoysala).
- The content shall be formatted with section headings, body text, and relevant imagery.

**FR-7: Artist Profile**

- The system shall display a profile screen for the featured Shilpi including name, location, experience years, specialisation, and biography.
- The profile shall display a professional photo of the artist at work.

## 8. Non-Functional Requirements

| Category      | Requirement        | Acceptance Criterion                                                                                                                         |
|---------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Performance   | Gallery Load Time  | All 20 gallery thumbnails must render within 2 seconds of home screen launch on a mid-range device with Wi-Fi or 4G.                         |
| Performance   | Image Transition   | Transition from thumbnail to full-screen viewer must complete within 300 ms.                                                                 |
| Performance   | Zoom Response      | Pinch-to-zoom must respond within one frame (approx. 16 ms) with no visible stutter.                                                         |
| Usability     | Inquiry Flow       | A buyer must be able to send a WhatsApp inquiry within 2 taps from any artwork detail screen.                                                |
| Usability     | Legibility         | All primary text must use a minimum font size of 14sp. Artwork titles must use a minimum of 18sp.                                            |
| Aesthetics    | Gallery Grade UI   | The UI must achieve a minimalist, premium art gallery aesthetic: white backgrounds, generous padding, and artwork as the visual centrepiece. |
| Reliability   | Image Integrity    | Glide must cache all gallery images locally so they remain viewable after initial load even with intermittent connectivity.                  |
| Compatibility | Android Versions   | Application must support Android 5.0 (API 21) through Android 14 (API 34).                                                                   |
| Compatibility | Screen Sizes       | The gallery grid layout must adapt correctly to screen widths from 360dp (budget phones) to 480dp+ (premium devices).                        |
| Privacy       | No Data Collection | The application must not collect, transmit, or store any personal data. All inquiry data is handled natively by WhatsApp.                    |

## 9. System Architecture

Shilpa-Kala Showcase follows a single-tier, client-only architecture. All image assets, artwork metadata, heritage content, and artist profile data are bundled within the application package (APK). There are no backend servers, cloud databases, or API dependencies required for the core gallery and browsing operation. The only external service dependency is WhatsApp, invoked via Android's standard Intent system for buyer inquiries.

### 9.1 Technology Stack

| Layer           | Technology                   | Purpose                                                                                                                            |
|-----------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Language        | Kotlin                       | Primary development language; concise, null-safe, fully interoperable with Android SDK.                                            |
| UI Framework    | XML Layouts + AppCompat      | Traditional View-based UI with RecyclerView, ViewPager2, and Material Design components for rich gallery interactions.             |
| Image Loading   | Glide 4.x                    | Smooth high-resolution image loading with memory and disk caching, placeholder support, and bitmap pooling for 20+ images.         |
| Zoom Library    | PhotoView / TouchImageView   | Gesture-based pinch-to-zoom and double-tap zoom on full-screen artwork images. Supports drag-to-pan beyond screen edges.           |
| WhatsApp Intent | Android Intent (ACTION_VIEW) | Launches WhatsApp with a pre-composed message URI containing product ID and artwork details. No WhatsApp SDK required.             |
| GenAI Content   | Static AI-Generated Text     | Heritage story and artwork descriptions generated via GenAI and bundled as static JSON content within the app for offline viewing. |
| Data Storage    | JSON / Assets Folder         | Artwork metadata stored as structured JSON in the assets folder. Parsed at runtime into Kotlin data classes.                       |
| Build System    | Gradle (Groovy DSL)          | Dependency management, build configuration, and APK generation.                                                                    |
| Min. SDK        | API Level 21                 | Targets approximately 98% of active Android devices globally, including affordable smartphones used by rural artisans.             |

### 9.2 Architectural Overview

The application follows a simplified MVC (Model-View-Controller) pattern appropriate for a content-rich Android gallery application:

- **Model Layer:** JSON data files in the assets/ folder define the artwork catalogue (title, artist, dimensions, material, product ID, image filename, WhatsApp number) and heritage content. A DataManager utility class parses and serves this data to activities.
- **View Layer:** XML layout files define the UI structure: activity\_home.xml (gallery grid), activity\_artwork\_detail.xml (detail + inquiry), activity\_timeline.xml (work-in-progress), activity\_heritage.xml (heritage story), activity\_artist\_profile.xml (Shilpi profile).

- **Controller Layer:** Activity classes orchestrate user interaction. HomeActivity loads the gallery grid via RecyclerView + GalleryAdapter. ArtworkDetailActivity handles the WhatsApp Intent. TimelineActivity renders the creation timeline. HeritageActivity presents AI-generated cultural narrative.
- **Image Pipeline:** Glide is configured with a custom RequestOptions instance specifying placeholder, error drawables, centerCrop transformation, and DiskCacheStrategy.AUTOMATIC. All high-resolution images are sourced from the res/drawable/ folders organised by sculpture category.

### 9.3 Artwork Catalogue Data Schema

| Field       | Type    | Description                                                                                        |
|-------------|---------|----------------------------------------------------------------------------------------------------|
| productId   | String  | Unique identifier per artwork (e.g., SKS-2024-001). Included verbatim in WhatsApp inquiry message. |
| title       | String  | Display name of the sculpture (e.g., Dancing Nataraja – Granite).                                  |
| artist      | String  | Name of the Shilpi who created this work.                                                          |
| material    | String  | Stone or wood type (e.g., Black Granite, Mysore Sandstone, Rosewood).                              |
| dimensions  | String  | Height x Width x Depth in centimetres.                                                             |
| style       | String  | Carving style category (Hoysala / Dravidian / Mysore / Contemporary).                              |
| description | String  | AI-generated description of the artwork for display in the detail screen.                          |
| imageResId  | Int     | Drawable resource ID for the high-resolution image asset.                                          |
| whatsappNum | String  | Artisan WhatsApp number in international format for inquiry routing.                               |
| inProgress  | Boolean | Flag indicating whether this artwork has an associated work-in-progress timeline.                  |

## 10. User Flow

The application is designed to deliver a gallery-browsing experience with minimal friction. The complete user journey from launch to WhatsApp inquiry is described below:

| Step | Screen                    | User Action and System Response                                                                                                                                                                                                                 |
|------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Home – Gallery Grid       | App launches directly to the gallery home screen. A RecyclerView grid displays thumbnail cards for all artworks. Category filter chips allow browsing by style (All / Deities / Panels / Architectural / Commissions).                          |
| 2    | Artwork Detail            | User taps any thumbnail. ArtworkDetailActivity opens with a large hero image, artwork title, material, dimensions, style, and AI-generated description. The Enquire via WhatsApp button is prominent at the bottom.                             |
| 3    | Zoom and Inspect          | User pinches or double-taps the hero image. Full-screen zoom viewer opens. User can pan and zoom up to 3x to inspect intricate carving details.                                                                                                 |
| 4    | WhatsApp Inquiry          | User taps Enquire via WhatsApp. Android launches WhatsApp with a pre-composed message: "Hello, I am interested in [Title] (Product ID: [SKS-XXXX]). Please share availability and pricing." The artisan receives this on their WhatsApp number. |
| 5    | Work-in-Progress Timeline | User taps the Creation Journey button on a detail screen. TimelineActivity displays a vertical scroll of photographs from stone selection to finished sculpture, with stage captions.                                                           |
| 6    | Heritage Story            | User navigates to Heritage from the bottom navigation bar. HeritageActivity displays AI-generated narrative about a carving style (e.g., Hoysala – 12th Century Karnataka) with section headings and imagery.                                   |
| 7    | Artist Profile            | User taps the artist name on any detail screen. ArtistProfileActivity shows the Shilpi's photo, biography, location, years of experience, and specialisation areas.                                                                             |

## 11. Technical Implementation

### 11.1 Glide Image Loading and Performance

All sculpture photographs are loaded using Glide 4.x, configured with a global RequestOptions instance applied via GlideApp. Thumbnails in the RecyclerView grid use a 480x480px downsampled version to minimise memory footprint, while full-screen detail views load the original resolution asset. Glide's DiskCacheStrategy.AUTOMATIC caches both transformed thumbnails and original images locally, ensuring fast re-display without re-downloading. The application is verified to handle 20+ concurrent high-resolution images without triggering OutOfMemoryError events by relying on Glide's internal BitmapPool for bitmap recycling.

### 11.2 Zoom Interaction — TouchImageView / PhotoView

The full-screen artwork viewer is implemented using TouchImageView (or the PhotoView library), which extends Android's ImageView to support multi-touch pinch-to-zoom, double-tap zoom toggle, and momentum-based panning. The minimum zoom scale is set to 1.0f (fit-to-screen) and the maximum to 3.0f. Double-tap cycles between 1.0f and 2.0f. The zoom view is wrapped in a ViewPager2 to allow swiping between artworks in full-screen mode while preserving zoom state per page.

### 11.3 WhatsApp Intent Integration

WhatsApp inquiry is triggered via Android's standard Intent mechanism using the URI scheme "https://api.whatsapp.com/send?phone=XXXX&text=ENCODED\_MESSAGE". The message is pre-composed in ArtworkDetailActivity using the artwork's productId, title, and the artisan's whatsappNum from the JSON catalogue. The text is URL-encoded using Uri.encode() before being embedded in the Intent URI. The Intent is dispatched with startActivity(), and a fallback dialog is shown if WhatsApp is not installed on the buyer's device.

### 11.4 RecyclerView Gallery Adapter

GalleryAdapter extends RecyclerView.Adapter<GalleryAdapter.ArtworkViewHolder> and binds each ArtworkItem from the parsed JSON catalogue to a card view. The adapter uses DiffUtil.calculateDiff() for efficient updates when category filter chips are changed, avoiding full adapter invalidation. Each card displays a Glide-loaded thumbnail, the artwork title, material, and a subtle product ID label in the corner. GridLayoutManager with a span count of 2 provides the gallery grid layout, dynamically adjusting item widths to fill the screen.

### 11.5 Heritage Content — GenAI Integration

Heritage story content is generated using a GenAI large language model prompted with historical references about specific Indian carving traditions. The generated content covers the origin period, geographical distribution, stylistic characteristics, notable temples and monuments, technique descriptions, and the current state of the craft. This content is reviewed, curated, and bundled as static strings in the heritage\_content.json asset file. HeritageActivity parses and displays this content with custom typography styling that evokes the feel of a museum guide.

## 11.6 Work-in-Progress Timeline

TimelineActivity implements a vertical timeline using a RecyclerView with a custom LinearLayoutManager and a TimelineAdapter. Each timeline entry (TimelineStage data class) contains a stage number, photograph resource ID, stage label, date/day marker, and caption text. The adapter draws a vertical connector line between entries using a custom ItemDecoration, creating the visual timeline effect without requiring a third-party library. Images within the timeline support the same touch-zoom capability as the main gallery viewer.

## 12. Project File Structure

| File / Directory                              | Purpose                                                                                                                              |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>app/build.gradle</b>                       | App-level Gradle config: PhotoView/TouchImageView, Material, Glide, Design dependencies.                                             |
| <b>data/ArtworkItem.kt</b>                    | Kotlin data class representing a single sculpture entry from the JSON catalogue.                                                     |
| <b>data/HeritageContent.kt</b>                | Data class representing a heritage story entry (style name, origin, history sections, imageResIds).                                  |
| <b>data/TimelineStage.kt</b>                  | Data class representing one stage in the work-in-progress timeline (stageNumber, label, dayMarker, imageResId, caption).             |
| <b>data/DataManager.kt</b>                    | Utility object that parses artwork_catalogue.json and heritage_content.json from assets/ and provides in-memory lists to activities. |
| <b>HomeActivity.kt</b>                        | Main launcher activity: hosts category filter chips and RecyclerView gallery grid. Handles category filtering.                       |
| <b>ArtworkDetailActivity.kt</b>               | Detail screen: loads full artwork data, displays hero image with zoom, and handles WhatsApp Intent on Enquire button tap.            |
| <b>TimelineActivity.kt</b>                    | Work-in-progress timeline screen: renders RecyclerView with custom timeline decorator and per-image zoom support.                    |
| <b>HeritageActivity.kt</b>                    | Heritage story screen: parses and displays AI-generated cultural narrative with styled typography and section imagery.               |
| <b>ArtistProfileActivity.kt</b>               | Artist profile screen: displays Shilpi photo, name, location, biography, and specialisation areas.                                   |
| <b>GalleryAdapter.kt</b>                      | RecyclerView.Adapter for the home gallery grid. Uses Glide for thumbnails and DiffUtil for efficient filtering updates.              |
| <b>TimelineAdapter.kt</b>                     | RecyclerView.Adapter for the timeline screen. Binds TimelineStage entries with stage labels and TouchImageView instances.            |
| <b>res/layout/activity_home.xml</b>           | Home screen layout: ChipGroup filter bar and RecyclerView grid.                                                                      |
| <b>res/layout/activity_artwork_detail.xml</b> | Detail screen layout: hero image (TouchImageView), info card, and Enquire button.                                                    |
| <b>res/layout/activity_timeline.xml</b>       | Timeline layout: RecyclerView with custom vertical connector decoration.                                                             |
| <b>res/layout/activity_heritage.xml</b>       | Heritage screen layout: ScrollView with styled text sections and embedded images.                                                    |
| <b>res/layout/item_gallery_card.xml</b>       | Gallery grid item: card with thumbnail ImageView, title, material, and product ID label.                                             |



| File / Directory                     | Purpose                                                                                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>assets/artwork_catalogue.json</b> | JSON array of all ArtworkItem records. Single source of truth for gallery content.                                                     |
| <b>assets/heritage_content.json</b>  | JSON array of HeritageContent records. AI-generated narratives per carving style.                                                      |
| <b>res/drawable/</b>                 | All high-resolution sculpture photographs organised by category. Also contains shape drawables for card backgrounds and button styles. |
| <b>AndroidManifest.xml</b>           | App manifest: permissions (QUERY_ALL_PACKAGES for WhatsApp Intent), activity declarations, launcher configuration.                     |

## 13. Permissions and Privacy

| Permission                | Required For               | Notes                                                                                                                                                                   |
|---------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>QUERY_ALL_PACKAGES</b> | WhatsApp Intent            | Required on Android 11+ to check whether WhatsApp is installed before dispatching the inquiry Intent. Used solely to provide a graceful fallback if WhatsApp is absent. |
| <b>INTERNET</b>           | Future features      cloud | Declared for future versions that may load images from a remote CDN. Not required for the current bundled-asset version of the application.                             |

No personal user data is collected, stored, or transmitted by Shilpa-Kala Showcase. The application does not use analytics SDKs, crash reporting services, or advertising networks. All gallery content (images, metadata, heritage text) is bundled within the APK. The only data exchanged with an external party is the buyer-initiated WhatsApp message, which is composed within the app but transmitted and handled entirely by the WhatsApp application. Shilpa-Kala Showcase has no visibility into or control over this message after it is dispatched.

## 14. Impact Goals

### 14.1 Artisanal Empowerment

By giving traditional stone and wood carvers a professional digital portfolio accessible from any smartphone, Shilpa-Kala Showcase opens a direct channel to national and international buyers who would otherwise never discover these artisans. Eliminating the intermediary layer allows craftsmen to capture significantly more of the value their art commands in the marketplace, improving their livelihoods and making the craft economically sustainable.

### 14.2 Export Potential — Make in India

India's handmade sculpture market has significant untapped export potential, with buyers in the United States, Europe, Southeast Asia, and the Indian diaspora actively seeking authentic traditional works. Shilpa-Kala Showcase positions every featured Shilpi as a Made in India brand, enabling high-value craft exports that align with the Government of India's Make in India and Vocal for Local initiatives.

### 14.3 Heritage Preservation

The AI-generated heritage story sections serve as a living digital archive of India's sculptural traditions. Techniques, style characteristics, historical contexts, and the names of master craftsmen are documented in a format that survives beyond the lifetime of any individual artisan. This digital record contributes to the preservation of intangible cultural heritage accessible to researchers, educators, and future generations of craftsmen.

### 14.4 Youth Re-engagement with Traditional Crafts

One of the root causes of craft decline is the perception that traditional carving has no modern career path. By demonstrating that a Shilpi's work can be showcased on a smartphone with the same professionalism as a contemporary art gallery, Shilpa-Kala Showcase helps shift this perception. A younger generation of craftsmen who see their art valued, inquired about, and purchased through a digital platform are more likely to continue the tradition.

## 15. Success Criteria

The project will be considered successfully completed when the following criteria are met:

| #  | Success Criterion                                                                                                                   | Measurement                                                         | Priority |
|----|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|----------|
| 1  | App handles 20+ high-resolution images without frame drops, memory warnings, or crashes.                                            | Glide performance profiling with Android Studio Memory Profiler.    | Critical |
| 2  | Full-screen image viewer supports smooth pinch-to-zoom up to 3x with no visible stutter on a reference mid-range device.            | Manual interaction test on Redmi 9A or equivalent.                  | Critical |
| 3  | Tapping Enquire on any artwork launches WhatsApp with a pre-composed message containing the correct product ID and artwork details. | Manual WhatsApp dispatch verification for all catalogue items.      | Critical |
| 4  | The WhatsApp message includes the product ID exactly as defined in the artwork catalogue JSON — no modification or truncation.      | Automated string comparison test for all 20 artwork entries.        | Critical |
| 5  | Work-in-Progress timeline displays a minimum of 5 creation stages for at least one sculpture, all images loading correctly.         | Manual walkthrough of timeline screen for all in-progress artworks. | Critical |
| 6  | Heritage story section displays a complete AI-generated narrative for at least one carving style with correct section structure.    | Manual content verification against source JSON.                    | High     |
| 7  | Gallery grid loads and is interactive within 2 seconds on a Wi-Fi or 4G connection.                                                 | Timing test with Android Studio Network Profiler.                   | High     |
| 8  | UI achieves minimalist gallery aesthetic: white background, generous padding, artwork-centred layout, no visual clutter.            | UI review against gallery aesthetic checklist by evaluator.         | High     |
| 9  | App does not crash during a complete walkthrough of all screens, gestures, and the WhatsApp inquiry flow.                           | Manual regression test covering all screens and interactions.       | High     |
| 10 | Application installs and runs correctly on Android 5.0 (API 21) and Android 14 (API 34).                                            | Cross-version installation and walkthrough test.                    | Medium   |

## Conclusion

Shilpa-Kala Showcase represents a focused and socially meaningful application of Android development and GenAI in the domain of cultural heritage preservation and artisanal empowerment. By addressing a clear and well-defined problem — the complete digital invisibility of India's master stone and wood carvers — the project delivers measurable practical value alongside technical merit.

The application's architecture prioritises gallery-grade visual quality, frictionless buyer inquiry, and offline content availability above all else. Every design decision — from Glide for high-performance image loading, to WhatsApp Intents for zero-friction inquiry, to GenAI-generated heritage narratives for cultural documentation — is motivated by a single question: will this showcase the artisan's work with the dignity and professionalism it deserves, and will it make it effortlessly easy for a buyer anywhere in the world to make contact?

The project also establishes a strong technical foundation for future expansion. The JSON-based catalogue architecture supports rapid addition of new artisans, artworks, and carving styles without code changes. The modular activity structure supports the addition of video documentation, AR preview, multi-language support, and payment integration as natural extensions of the existing codebase.

In conclusion, Shilpa-Kala Showcase is not simply a technical deliverable — it is a bridge between ancient craft and the modern world, a small but meaningful step toward ensuring that the extraordinary artistic heritage of India's stone and wood carving traditions survives, thrives, and finds its rightful place on the global stage.

— *End of Report* —

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